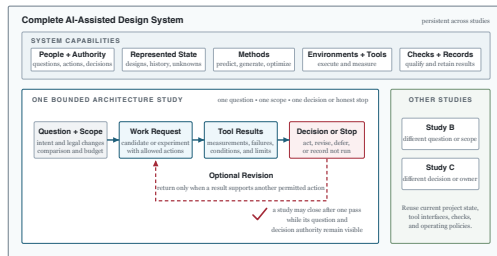


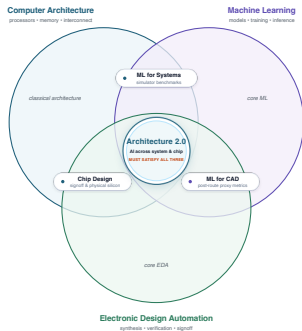
Architecture 2.0

Principles of AI-Native System and Chip Design



Three fields, one architecture decision

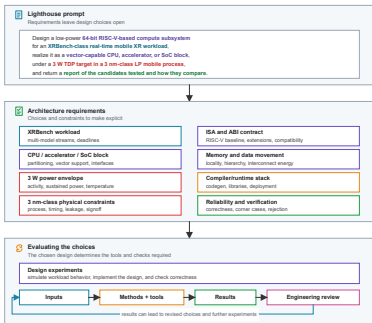
The useful intersection is not AI applied to chips. It is an architecture result that can support a decision.



- Architecture supplies the tradeoff and system boundary.
- ML supplies possible prediction, generation, and optimization methods.
- EDA supplies executable constraints and physical observations.

The Moonshot: from intent to a supported design

What if an AI-assisted design system could turn a compact architecture intent into an implementable, verified, and reviewable system design?



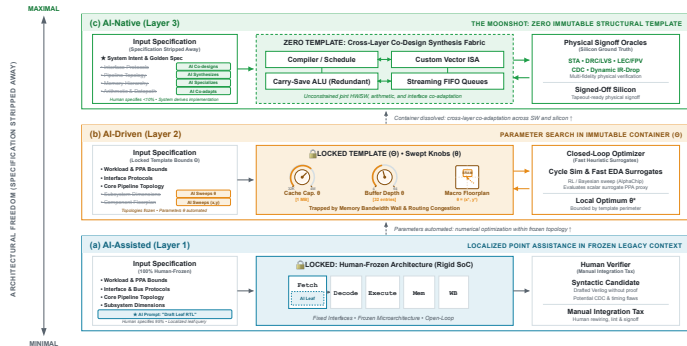
A field-scale target

- low-power, 64-bit RISC-V compute subsystem
- real-time mobile extended-reality (XR) workloads
- software, RTL, physical, verification, and product constraints

Source: Architecture 2.0, Chapter 1; XRBench and ILLIXR ground the workload class.

Three operational layers of AI integration

Assistance divides into point assistance in a frozen context, parameter search in an immutable container, and cross-layer co-design under physical signoff.



Layer 1: AI-Assisted

- Point syntax in frozen RTL
- Open-loop human review

Layer 2: AI-Driven

- Parameter search $\theta \in \Theta$
- Locked container; surrogates

Layer 3: AI-Native

- Zero structural template
- Closed-loop physical signoff

Source: Architecture 2.0, Chapter 1; Section 1.3.

Four different success claims

A candidate, a supported architecture result, the contribution of the complete AI-assisted approach, and moonshot capability are distinct claims.

Candidate Artifact

Legal artifact exists
under syntax rules

Supported Result

Evidence backs
the proposal

Method Contribution

Matched ablation
proves AI value

Moonshot Capability

Repeated trials
span declared tasks

None of these claims authorizes architecture commitment without physical signoff.

The Architecture 2.0 bet

Improve the architecture result, reduce the total cost of reaching a comparable result, or both.

Architecture quality

- performance, power, area
- reliability, security
- software and product fit

Total cost

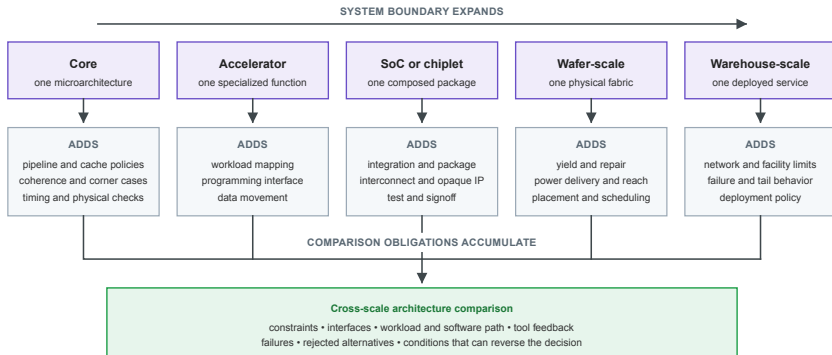
- model and tool calls
- queue and license time
- human setup, review, repair

Failure behavior

- invalid candidates
- missed constraints
- unsafe or uncontrolled actions

Every efficiency lever adds obligations

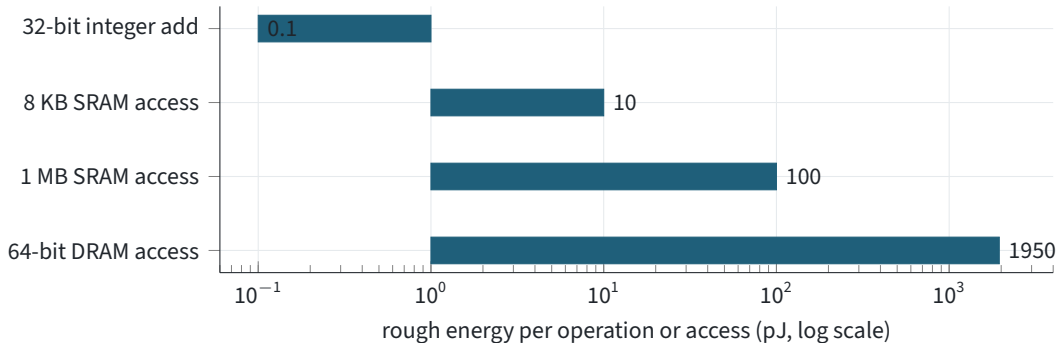
Scaling gave way to specialization, composition, and cross-layer work.



Source: Dennard et al. (1974); Amdahl (1967); Esmaeilzadeh et al. (2011); UCle Consortium (2026); Barroso, Hölzle, and Ranganathan (2019). Synthesis in Chapter 2.

Data movement can dwarf arithmetic

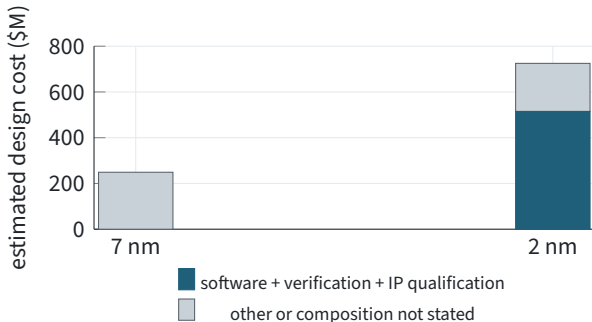
The system boundary decides whether a local compute gain matters.



Source: Horowitz, ISSCC 2014 rough 45 nm energy anchors. DRAM is plotted at the midpoint of the reported 1300–2600 pJ range; these are order-of-magnitude anchors, not current-node predictions.

Cost has moved beyond drawing the RTL

At advanced nodes, software, verification, and IP qualification can dominate estimated design cost.



71%

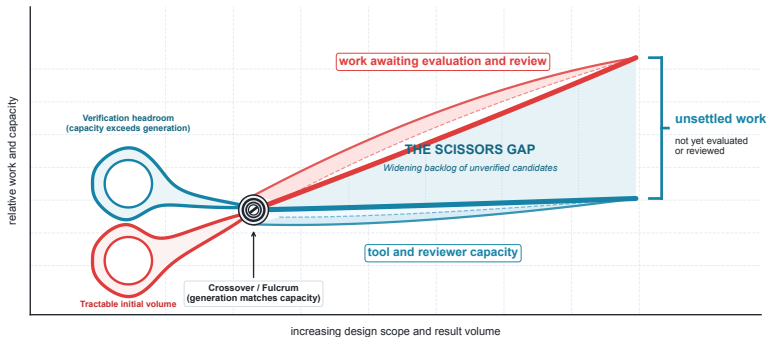
of the \$725M 2 nm estimate was attributed to the combined software, verification, and IP-qualification categories.

The 7 nm composition was not published, so the chart does not invent it.

Source: Arm Holdings, Form 424B4 (2023), reporting July 2022 IBS estimates: \$249M at 7 nm and \$725M at 2 nm.

Production can outpace examination

Faster candidate creation is useful only if evaluation and review can keep up.



This is one possible limiting pattern; the next slide diagnoses other cases.

Diagnose the limiting work first

Different delays call for different interventions.

Missing information	retrieve, reconcile, or collect the state
----------------------------	---

Invalid or unhelpful choices	constrain or change the candidate space
-------------------------------------	---

Expensive evaluation	predict, screen, or allocate stronger checks
-----------------------------	--

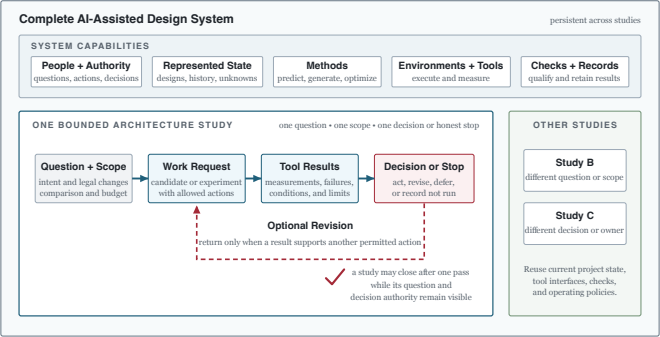
Overloaded review	change capacity, routing, or review preparation
--------------------------	---

Unclear decision	name the action, owner, and evidence threshold
-------------------------	--

“Use AI” is not a diagnosis.

A foundation model supports a larger design system

A foundation model can supply reusable architectural knowledge and capabilities. The complete design system still orchestrates people, methods, tools, project state, and checks.



System, study, and loop are different

The loop is a possible behavior inside a study, not the whole thesis.

Complete system

- persistent people, methods, tools, state, checks
- supports many studies

Architecture study

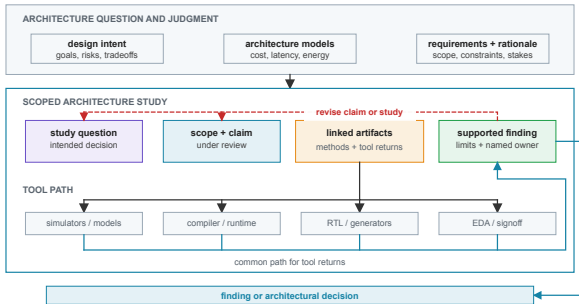
- one question and scope
- one decision or honest stop

Optional loop

- another permitted action
- only when the result supports it

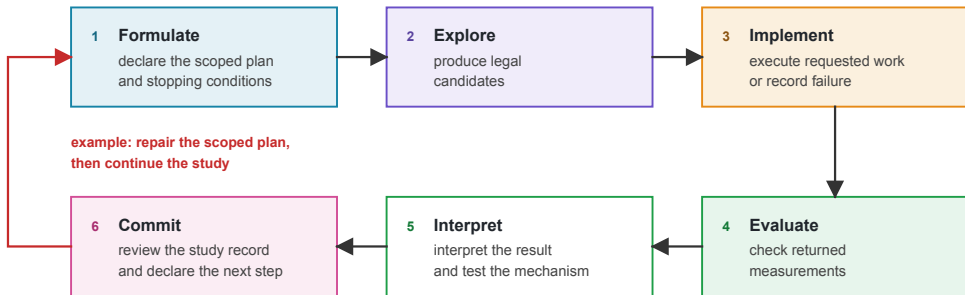
An architecture study connects question to decision

The study preserves the question, comparison, execution conditions, checks, and decision across tools.



Six distinct kinds of work

The stages are separated because they answer different questions and fail differently.



Repair the part that failed

Bad question

reformulate

Weak comparison

change alternatives or budget

Broken execution

repair the environment

Inadequate check

strengthen measurement

Failed explanation

narrow or retest mechanism

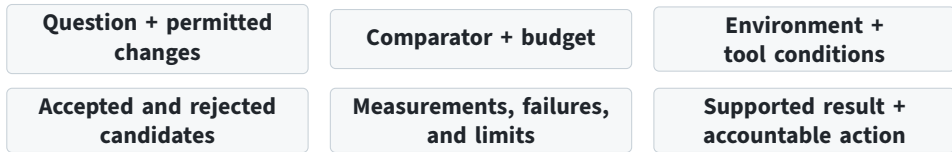
Unresolved decision

defer, reject, or assign work

Iteration should have a technical reason and a destination.

One record holds the comparison together

Another architect should be able to reconstruct what changed, what ran, what failed, and what the result supports.



The record connects the technical argument; it is not the argument itself.

The project knows three different things

General knowledge mechanisms, specifications, papers, slow-moving relationships

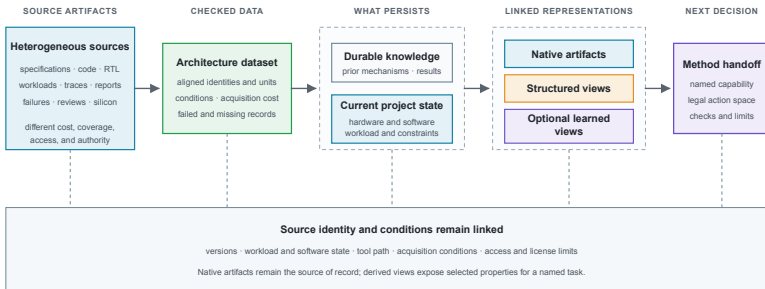
Current project state exact design, workload, software, constraints, owners

Project history runs, failures, rejections, adoption, changed conditions

A model's general knowledge is not live project state.

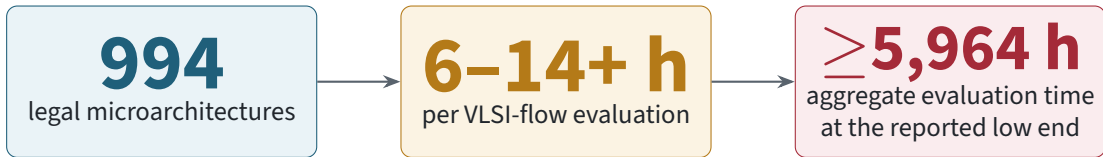
Observation sources are not interchangeable

Choose the source by the property the study must observe, not by a single fidelity ranking.



An architecture sample can be expensive

Data acquisition is part of the architecture method, not background plumbing.

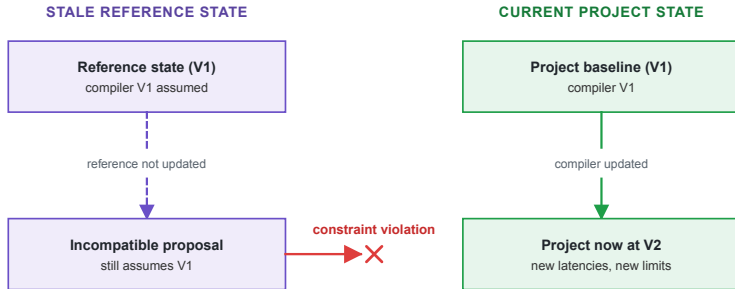


Ten analytical calls, ten cycle simulations, and ten physical-design runs are not ten equivalent samples.

Source: BOOM-Explorer: Bai et al., ICCAD 2021. The final value is $994 \times 6 \text{ h}$: a serial aggregate, not elapsed wall time.

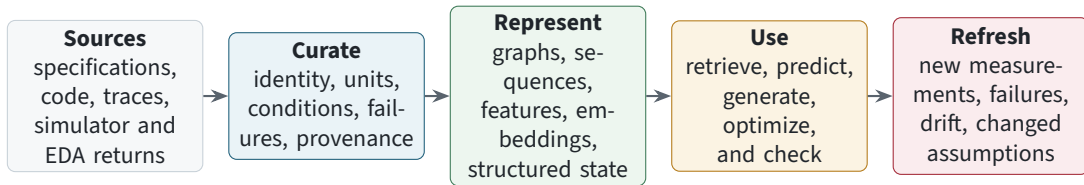
Representation determines reachable designs

A legal proposal can still be invalid when project state and tool-facing artifacts disagree.



Architecture data must become usable state

Data is made through tools and design work; representation determines what a model can learn and what the design system can act on.



An embedding is one learned representation, not a substitute for project identity, conditions, or tool-facing artifacts.

Role comes before method

Do not collapse an engineering job, a method family, and an implementation into “the AI.”

1. Engineering role propose, estimate, select, check, operate, interpret

2. Need for an added method direct tools or existing analysis may already suffice

3. Method family generation, prediction, optimization, or a combination

4. Implementation classical, learned, language-model based, or hybrid

Use the smallest sufficient approach

Choose the least costly approach that resolves the named bottleneck.

Direct analysis

closed form, roofline

Enumeration

when the space fits

Established tools

compiler, simulator, EDA

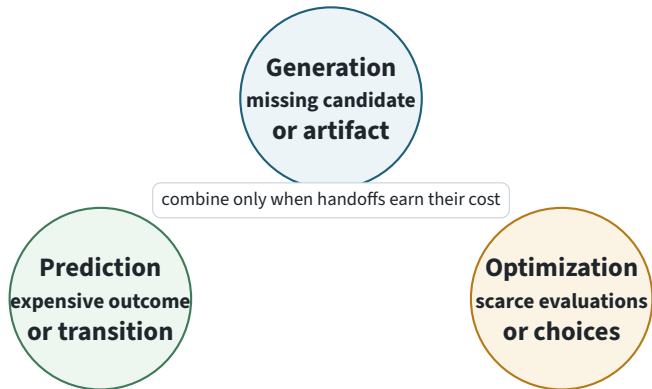
Added method

generator, predictor,
or optimizer

An added generator, predictor, or optimizer is optional when direct analysis or tools suffice.

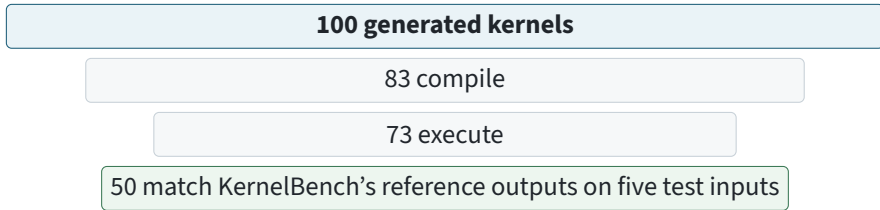
Three method families

Start from what is missing, not from a fixed prediction-generation-optimization sequence.



Generation expands; verification narrows

Attempted, executable, test-passing, and useful are different conditions.

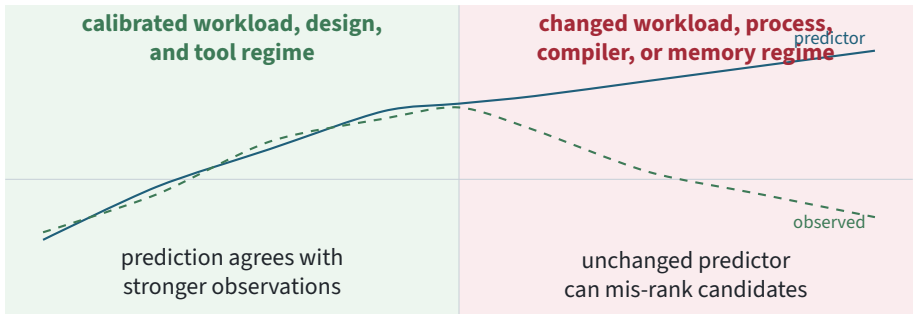


Passing five tests does not establish correctness outside those inputs.

Source: KernelBench, Ouyang et al. (2025), separate 100-kernel error-analysis sample. Per-model one-shot results in the paper show the same attempted → correct → correct-and-faster narrowing.

Prediction trades cost for a support assumption

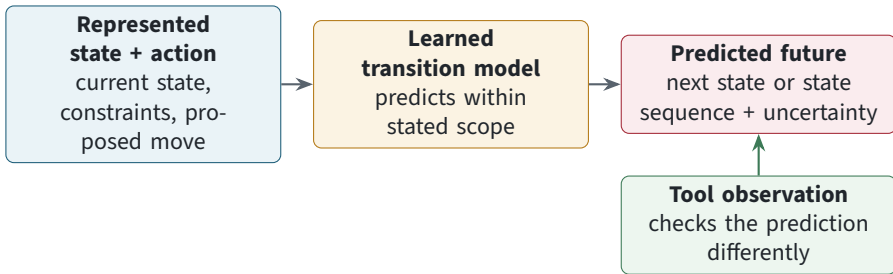
A cheap predictor helps only where its error is acceptable for the decision it screens.



Source: Constructed teaching illustration; Chapter 5 develops calibration and support limits.

A world model is a learned transition model

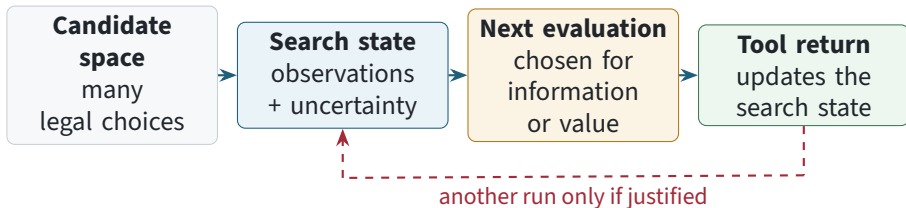
It predicts a sequence of future states from a represented state and proposed actions. Here it supports prediction or planning; it is not the live project record or a verification tool.



The observation can test model support; it does not make the model authoritative.

Optimization can allocate the next evaluation

Optimization selects or improves candidates under an objective, constraints, and a stopping rule. For costly black-box objectives, it must also decide what to measure next.



Method handoffs must earn their cost

This is one possible combination; no fixed order is implied.



Handoffs carry assumptions about representation, support, and budget. Tool returns still need qualification.

Justified

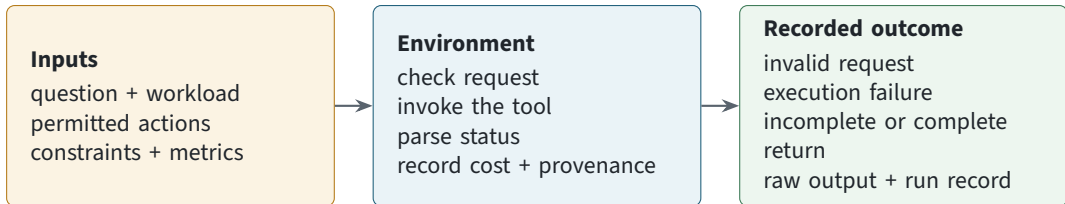
- each stage removes more cost than it adds
- rejected candidates and failures stay visible

Not justified

- more components without a named bottleneck
- a cheap proxy replaces the decisive check

An environment is more than a simulator

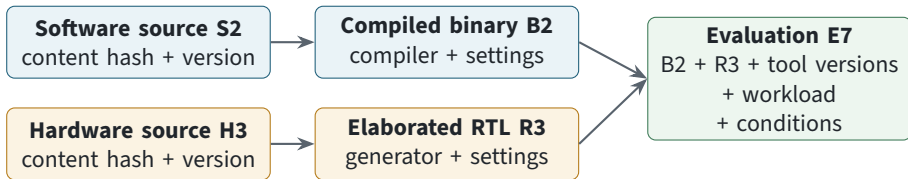
An environment defines how a design request becomes a recorded tool outcome.



Define permitted actions, fixed context, expected outputs, cost, lineage, and status before execution.

Identity must survive every tool

A successful command is not enough to reconstruct a result.



Artifact identity = content + production lineage.
**Evaluation identity = exact artifacts +
tool versions + workload + conditions.**

A return is not yet a measurement

A value becomes a measurement only when its identity, conditions, status, and limits are bound to it.



If qualification fails, retain an execution return, not a measurement for comparison.

Formal and simulation answer different questions

Formal

- proves a stated property or finds a counterexample
- under a model and assumptions
- may be inconclusive or time out

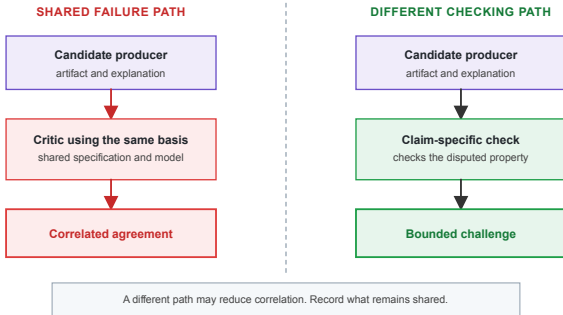
Simulation

- observes exercised behavior
- under selected stimuli and configuration
- cannot exhaust a modern design

Neither check repairs an incomplete requirement or system boundary.

Another model is not an independent check

Independence comes from a materially different observation path, not another fluent answer.



Feedback must update the right thing

An invalid execution return goes back to the environment. A qualified or verified result can direct feedback to the candidate, method support, project state, or reusable knowledge; each update receives an appropriate recheck.

Candidate repair

change the artifact
or configuration

Method support

recalibrate or narrow
supported use

Project state

update constraints, identity,
or tool status

Reusable learning

generalize only
after repeated
independent checks

Learn when justified

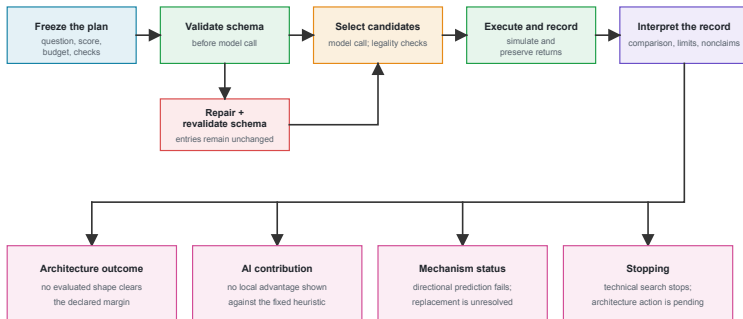
- enough qualified observations
- held-out or independent recheck

Do not silently learn

- one anomalous return
- changed workload or tool regime

One design loop separates four judgments

The prospective XR cache record stops before execution. A separate executed synthetic GEMM array study shows how a design loop separates outcome, contribution, mechanism, and stopping.



The executed array study produced a local tie

Frozen local score = total modeled cycles ÷ average layer utilization; lower is better. It is a study-local ordering rule, not a physical measurement.

Model arm	4 simulator calls	score 33,782	32×32 and 16×64 tied
------------------	-------------------	--------------	----------------------

Fixed heuristic	4 simulator calls	score 33,782	32×32 and 16×64 tied
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No evaluated nonbaseline cleared the margin.
Technical recommendation: retain 32×32 for the evaluated set and stop.
Accountable confirmation remains pending; no architecture commitment.

Post-run check: 16×64 had twice the baseline's modeled DRAM-word writes, omitted from the frozen score.

Source: Retained SCALE-Sim 3.0.0 study in Chapter 8. Synthetic three-layer GEMM, not an XR trace or Lighthouse result; no architecture commitment or silicon measurement.

The failed explanation is also a result

In this general matrix multiplication (GEMM) contrast, the local comparison survives even though the proposed explanation fails.

Setup

In a weight-stationary dataflow, weights stay in the array. K maps to array height, N to array width, and M streams through.

16×64 array
height 16, width 64

64×16 array
height 64, width 16

Matched workload contrast: swap M and N ; keep K unchanged.

Prediction

Transposing M and N would reverse the winner.

Observed

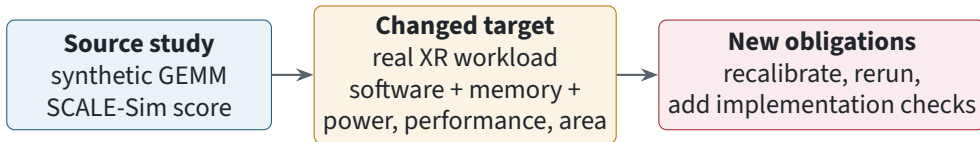
16×64 ranks ahead of 64×16 in this contrast.

**Keep the observed ordering.
Reject the proposed mechanism.**

Source: SCALE-Sim Project v3.0.0 source and retained Chapter 8 execution records. This synthetic GEMM contrast is separate from the prospective XR record.

A new decision reopens the claim

Study structure may transfer; a local conclusion does not transfer automatically.



Workload, software, hardware, tools, process assumptions, and decision scope are part of the result.

What transfers is structure, not the answer

Across design problems, reuse the disciplined parts of the study and re-establish the facts that changed.

Often reusable

- question and comparison template
- environment interface and run record
- method role and independent checks
- cost accounting and stopping logic

Must be re-established

- workload and software behavior
- model calibration and support
- tool, process, and physical conditions
- evidence threshold and decision authority

Transfer a method only as far as its assumptions and checks still hold.

Four patterns recur across the stack

The artifacts differ from software to circuits, but the architecture problems repeat.

Costly decisive evaluation

Use cheap bounds and predictors to decide which candidates deserve stronger tools.

Coupled choices

Hardware, software, tools, and workloads move together; preserve the complete comparison.

Partial observability

Represent conditions and uncertainty; do not treat an absent measurement as a favorable one.

Verification effort follows failure cost

Generate cheaply when possible, but scale checks with the cost of a wrong commitment.

Evaluate matched complete systems

Matched complete systems can estimate a workflow-package adoption effect. A model score cannot.

Same decision, starting information, evaluator-owned tools and checks, cost window, and stopping rule

AI-assisted design system

- internal models, methods, and tools
- operator expertise and intervention
- failures and all resource costs retained

Strong comparison system

- established heuristic or human process
- practitioner expertise and intervention
- failures and all resource costs retained

Compare paired outcome–cost results across declared budgets and repeats. Controlled replacement or ablation tests a component’s contribution, but interactions may leave attribution unresolved. Neither assessment authorizes architecture commitment.

Metrics people forget to count

Six metric groups support four portable judgments: architecture result; coordination and decision quality; total human and machine cost; and reliability across repeated, changed, and adversarial conditions.

Architecture performance, power, area, reliability, software and product constraints

Work performed candidates, model calls, tokens, tool calls by type, unique simulator or implementation runs

Resources + people wall/queue time, compute, energy, licenses, data preparation, training, setup, review, repair

Experimental validity task/workload coverage, paired repeats/seeds, dispersion, leakage/contamination checks

Evidence + decision qualification, uncertainty, check coverage, supported or unresolved status, reopening

Failures + control invalid actions, retries, timeouts, recovery, budget violations, unsafe changes

Ten tool calls do not have one common cost.

Stress-test failures; red-team attacks

Fault injection probes resilience. Red teaming predeclares the asset, adversary, access, attack goal, success criteria, containment, and recovery.

Artifact attack

hidden logic, backdoors, unsafe interfaces, physical side effects

Evaluator attack

weakened tests, proxy gaming, omitted conditions, contaminated data

Workflow faults

silent retries, hidden failures, uncontrolled tool or model spending

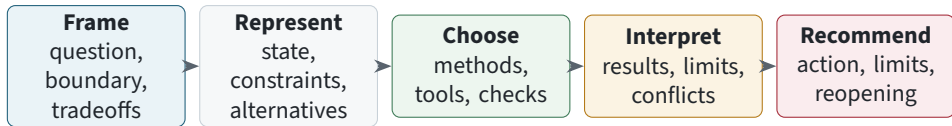
Authority attack

changed requirements, unauthorized waivers, recommendation treated as approval

Source: Chapter 10. Check final artifacts and downstream tool stages, not only source RTL; record attack success, detection, containment, recovery cost, and residual risk.

What remains architectural

The architect's role is a positive technical contribution, not whatever automation has not absorbed.



The architect produces a supported technical recommendation. Named commitment authority decides whether the design advances.

The field now has a research agenda

The next advances must improve design capability and the discipline for judging it.

Represent Build architecture data and state representations that preserve meaning across tools and time.

Compose Learn when prediction, generation, optimization, and classical analysis should cooperate.

Verify Develop independent checks that scale with generated candidates and physical consequence.

Evaluate Compare complete systems on architecture results, total cost, robustness, and decision quality.

Judge Establish what the result supports, state the technical recommendation, and name who can commit.

The goal is not autonomous output. It is better architecture work that can be defended.

Three judgments that remain separate

A tool return becomes a measurement only after its identity, conditions, status, and limits are bound to it.

Qualified measurements and checks support a technical recommendation

A matched evaluation separately asks whether the complete AI-assisted design

The named commitment authority decides whether a particular design

The recommendation and complete-approach assessment answer different questions.

Neither authorizes commitment, and the assessment is not a prerequisite for